PROBLEM → RESULT

Where is the race car? *Cameras only*, at race speed, with no start pose.

Causal x, y, yaw at 10 Hz from the car's cameras, wheel speeds and steering. No GNSS, no LiDAR, no starting pose given. A geo-referenced visual map, odometry and an EKF, all classical and explainable.

POSITION RMSE	LATERAL RMSE	HEADING RMSE
0.51 m	0.30 m	0.31 °

POSES WITHIN 1 M	1-S INTERVALS WITH A POSE	TIME PER FRAME · APPLE M4 CPU
95.1 %	99.8 %	24–39 ms / 100

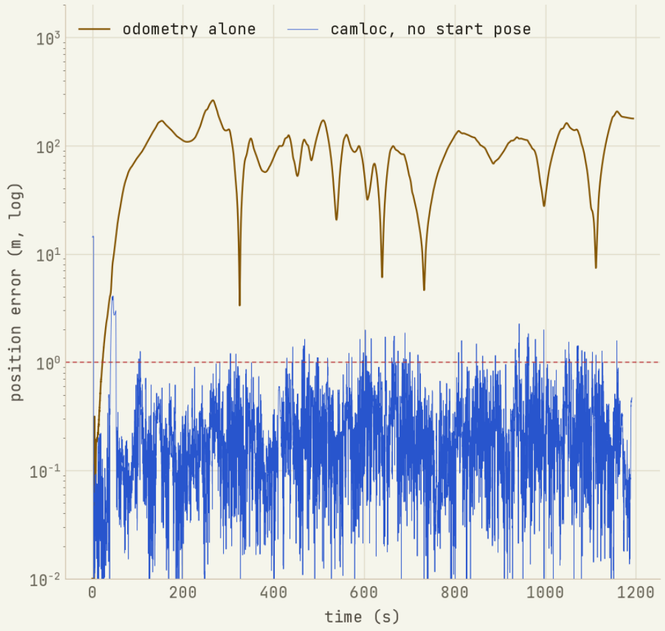
CLASSICAL · EXPLAINABLE | CAUSAL | CPU ONLY | NO PRE-TRAINED MODELS | NO EXTERNAL DATA | NO START POSE

Scope: held-out night laps 2 / 4 / 6 of the training run, pooled; the visual map and every calibration exclude the scored lap; no start pose. Coverage: full training run via run.sh (no pose during the first 3 s of search). Runtime: mean per frame, full run (24 ms) and 4 parallel folds (39 ms). Not a hidden-test result.

WHY NOT JUST ODOMETRY?

Wheels and steering alone drift *hundreds of metres*.



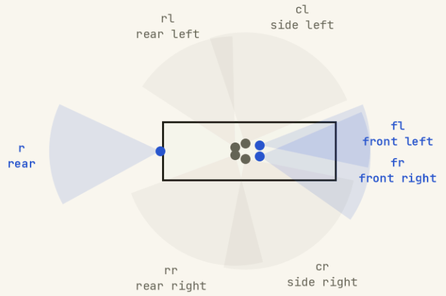


ODOMETRY ALONE · WHOLE RUN

107 m RMSE

up to 263 m off, even starting from the true pose

7 CAMERAS · 10 HZ · NIGHT



fl · fr · r carry the map fixes; the side cameras sit about 0.2 m above the paint, so lateral position from lines is unobservable.

Scope: training run, 1195 s, night. Odometry: calibrated wheel speeds + steering (bicycle model), started from the GT pose (display only). Ground truth is drawn for display only. Camera images are not shown (dataset licence: no redistribution).

PIPELINE · CAUSAL, EVERY FRAME

A known map, odometry and one filter. *No neural networks.*

INPUT
Causal MCAP replay: images grouped by stamp, never ahead of time

PREDICT
Wheel speeds + steering, braking-aware axle choice
`(or validation/twist)`

RELATIVE YAW
Keyframe rotation: LK flow + essential matrix

ABSOLUTE FIX
CLAHE ORB features → geo-referenced 3D map → PnP + RANSAC per camera, weighted by inliers and reprojection error

GATE
Cross-camera consensus, then a NIS test that recovers if the filter itself is wrong

EKF
x, y, yaw, keyframe yaw, online sideslip · track and pit-lane geometry

NO START POSE

Find yourself, *then check*.

Global search: descriptor votes over the whole map → PnP on the top 8

Provisional lock: 2 cameras agree, or 3 frames agree after odometry

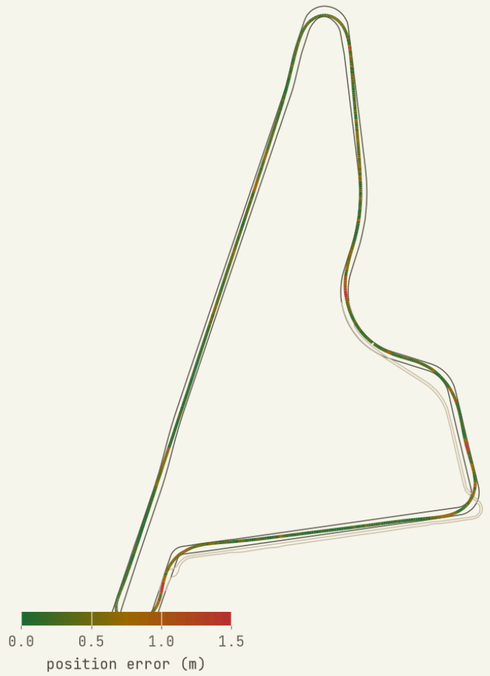
Confirm after 2 accepted fixes, or switch when ≥ 3 rejected fixes agree elsewhere

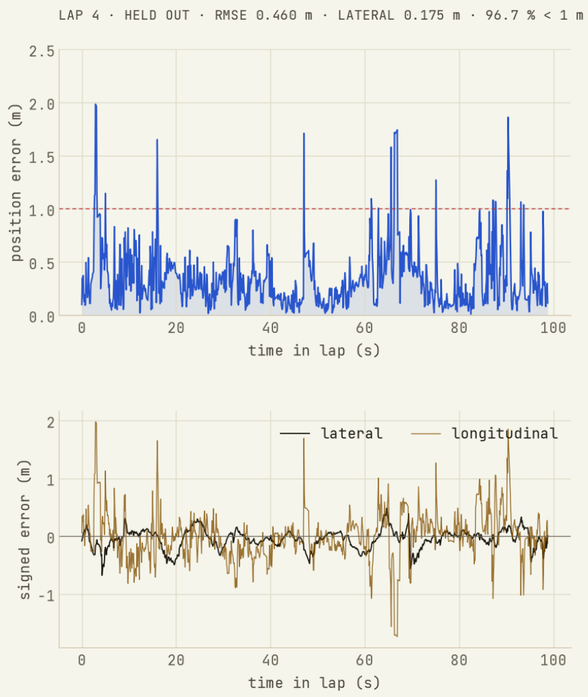
Time-boxed search; never silent after 3 s

Declarations: no pre-trained models, no external data. OpenCV ORB / LK / PnP only. Car signals used: wheel speeds, steering, brake pressure (allowed set, ADR-0003). The GT pose and /tf map→base_link are never read at inference; GT is used only for training-time calibration, the map and scoring.

RESULTS · HELD-OUT LAPS

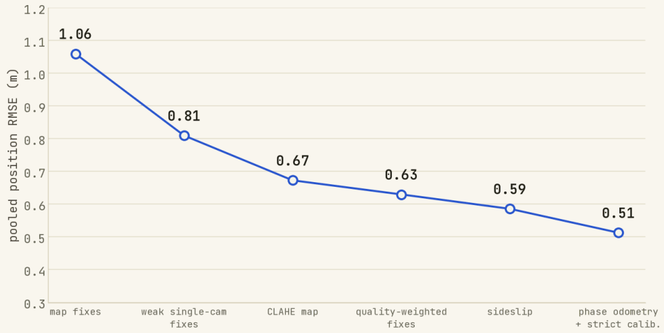
Sub-metre on every held-out lap, *from any start*.





HELD-OUT LAP	POS RMSE	LATERAL
Lap 2	0.598 m	0.404 m
Lap 4 (fastest)	0.460 m	0.175 m
Lap 6	0.415 m	0.206 m
Pooled	**0.513 m**	**0.303 m**

POOLED RMSE BY DEVELOPMENT STEP



16 MID-LAP STARTS, NO START POSE

16 / 16 end correct. Lock in 0.0–1.8 s. An aliased lock 13.6 m off **switches after 0.3 s**; one 9.9 m off converges by 11 s.

Scope: night training run, laps 2 / 4 / 6 each held out: the map and every calibration are rebuilt without the scored lap (strict). The progression points before the last one exclude only the scored lap from the map. Mid-lap starts: 40 s windows at 10–90 % of each held-out lap.

ROBUSTNESS · HONEST LIMITS

What we tested, and *where it breaks*.

STRESS TESTS

2 % of map fixes shifted 10 m	0.457 m
Twist-only input (test layout)	0.602 m
Measured front exposure	0.5-1 ms
Motion blur up to ~60 m/s	small
Mid-lap starts ending correct	16 / 16

Wrong-fix test: lap 4. Twist: pooled held-out laps. Blur at 60 m/s is extrapolated from the measured exposure.

LIMITS

30 s without map fixes	2.7 m
60 s without map fixes	5.2 m
Aliased start: time to correct	0.3-11 s
Daylight sample: accepted fixes	11 in 74 s
Lateral from paint (0.2 m height)	unobservable

Blackouts: lap 4, RMSE over the window. The daylight sample has no GT: a structural check only (locks at 0.6 s, 74 / 74 one-second bins). Daylight against a night map means sparse fixes, so more dead reckoning.

Scope: all numbers from the night training run with held-out laps, except the organisers' daylight sample (structural check, no ground truth, never used for tuning). Experiments EXP-0030-0039 in experiments/experiment_log.md.

RUN IT

One command, *both topic layouts*.

```
$ ./run.sh test.mcap out.csv
```

→ out.csv (timestamp_ns, x, y, yaw) + out.run.json (per-frame time, hardware). Builds the venv and the visual map on first use.

VALIDATION

- Strict leave-one-lap-out: map + calibration
- Causal processing, image stamps only
- GT pose never read at inference

ENGINEERING

- 26 automated tests
- Python 3.10–3.12, CPU only
- Training and test-domain layouts

TRACEABLE

- 39 logged experiments (EXP-####)
- Decisions as ADRs
- Independent reviewer loop

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TEAM: <NAMES — OWNER FILLS IN>

Scope: run.sh with no flags was checked on the full training run and on a 60 s test-domain cut (no state_estimation, /tf or track_bounds; speed and yaw rate from validation/twist).